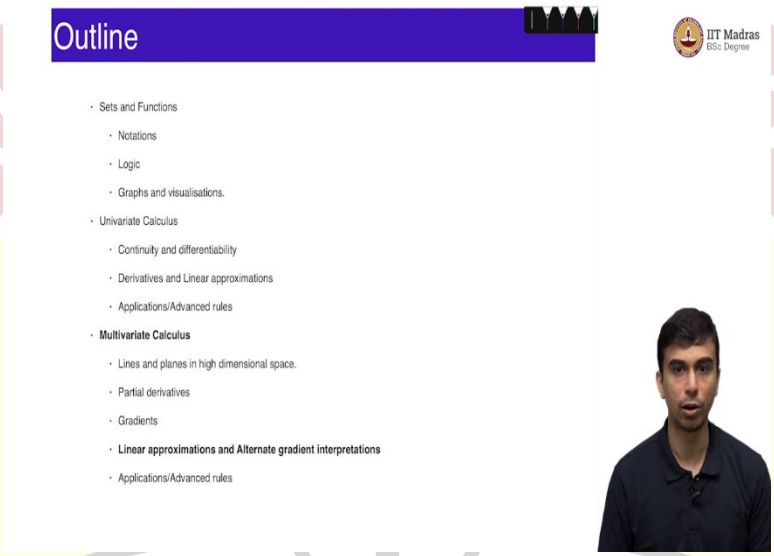


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
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Indian Institute of Technology, Madras
Multivariate Calculus: Linear approximation and applications

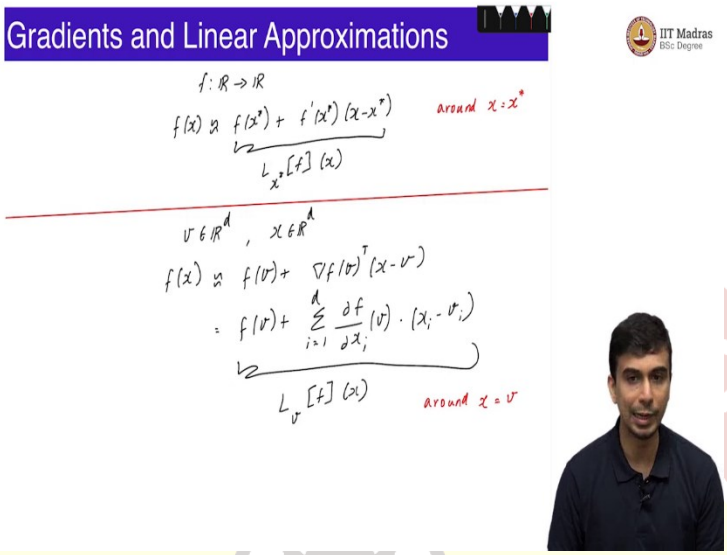
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Hello, everyone, and welcome to another lecture on machine learning foundations. In the last class, we introduced and define partial derivatives and gradients. And that is the first and standard interpretation of a gradient, which is it is a package containing the components, all the components of the partial derivatives. You have the partial derivatives with the respect of all variables and put them all together into one vector that is the gradient that is the computational interpretation of the gradient.

In today's lecture, we will take a look at the other interpretations of the gradient which are more geometric and more intuitive in nature than the computational approach.

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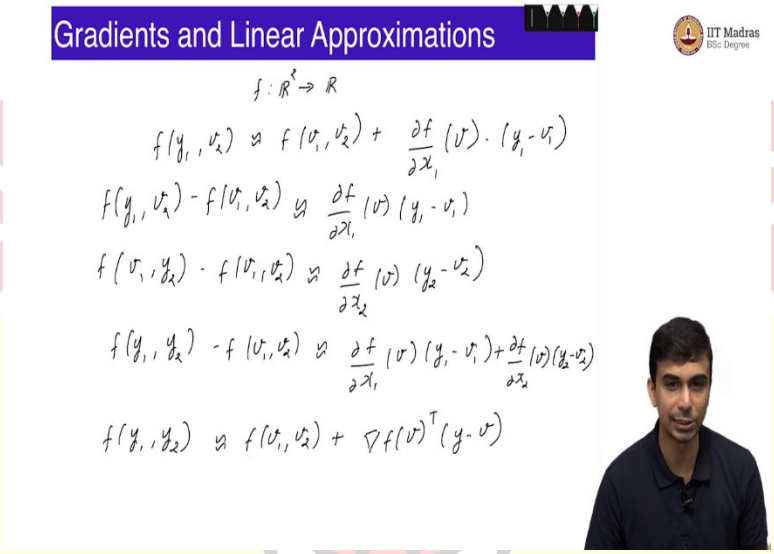
The first of which is the linear approximation interpretation. We already have seen linear approximations for one dimensional functions. If you have a function f from R to R, and you want to approximate this function f around x close to x star this is what you would do, f of x is approximately equal to f of x star + f dash of x star into x - x star. This is what you have been doing for linear approximations. And we denoted this term as the linear approximation of f around x star evaluated at x. And this is a valid approximation when x is close to x star. This is for one dimensional functions.

How can we do the same thing for functions from Rd to R? In other words, the it is, x, now imagine v is a vector, a d dimensional vector and x is in its neighborhood, x is a d dimensional vector and x is the close to v. Now, you want to be able to approximate x, the answer, I will give the answer first and then give various interpretations of it, which is f of x is approximately equal to f of v + grad f of v transpose x - v. You can see the similarities to the one dimensional case. So, f of x, you first have the f of v here, which corresponds to f of x star here, and you have grad f of v transpose x - v, which corresponds to f dash of x star into x - x star.

We will break this down even further as f of v + sum over i equal to 1 to d doh f by doh xi into xi - vi. I have just broken down the gradient into its components, doh f by doh xi evaluated at the point v multiplied by xi - vi. And we will denote this also in a very similar way. We will call this as the linear approximation of f around the point v evaluated at x.

And when is this approximation valid, this approximation is valid around x, approximately around x equal to v. When x is close to v, this is a valid approximation of f. So, we will give one interpretation or one possible interpretation of this lean approximation.

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The way we do that is by breaking down into multiple one dimensional functions. We will make life easier for ourselves and just consider a function, two dimensional function. This easily extends to multi dimensional functions as well, but we will do that for just two dimensional function. Let us say an f from R2 to R, f of x1 , v2 is approximately equal to f of v1 , v2.

We are viewing the second variable as a constant, the first variable as the variable. So, we are approximately around v1. So, f of x1 , v2 is approximately equal to f of v1 , v2 + doh f by doh x1 evaluated at v into x1 - v1. Here is a small notation issue. We will instead of calling it x1, we will call it y1. What is, this is just a standard one dimension linear approximation, that is we are approximating the function f as viewing as just as a function of its first argument, that is f of y1 , v2 is approximately equal f of v1 , v2. And we are using the partial derivative of f with respect to the first argument at v × y1 - v1.

We will simplify, we will change the notation a little bit more, f of y1 , v2 - f of v1 , v2 is approximately equal to doh f by doh x1 at v into y1 - v1. So, you can do the same thing for the second argument also, that is f of v1 , y2 - f of v1 , v2 this is approximately equal to doh f by doh x2, which is the second argument, at v into y2 - v2.

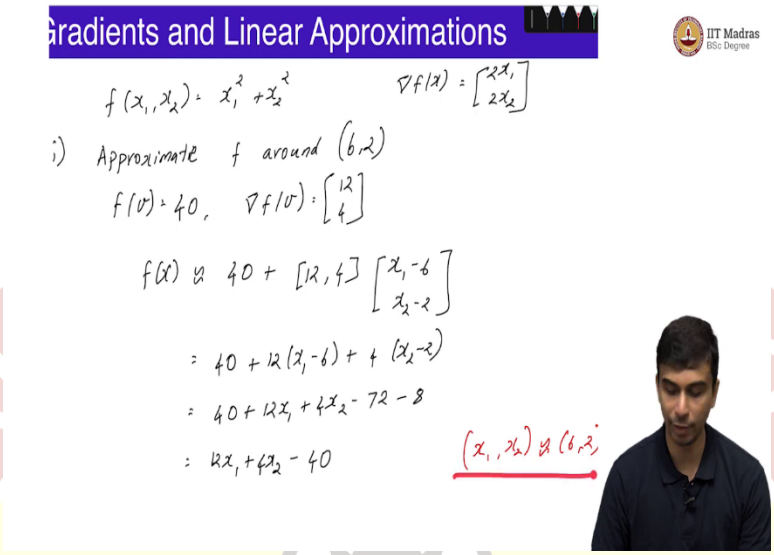
So, let us interpret these two approximations. What this says is, as I move from v1 , v2 to y1 , v2, that is as I move along the first axis, v1 changes to y1, v2 remains the same. This is the change in the value of the function. The value of the function changes by doh f by doh x1 evaluated at v into y1 - v1.

Similarly, here, this expression says that as I move from v1 , v2 to v1 , y2 the change is given by doh f by doh x2 evaluated at v into y2 - v2. But what we are now interested in is what if both of them move simultaneously? What if, what can I say about f of y1 , y2 - f of v1 , v2. The simplest answer is to, well, if one variable changes, this is the change, if another variable changes is the change. If both variables change, the change is simply given by the addition of these two changes, which is doh f by doh x1 at v into y1 - v1 + doh f by doh x2 at v into y2 - v2.

Or in other words, now we will write down the full linear approximation, which is f of y1 , y1 is approximately equal to f of v1 , v2 + doh f by doh x1 evaluated at v × y1 - v1 + doh f doh x2 evaluated at v × y2 - v2. Here, we are going to make our life easier. And I am just going to call it the gradient of f at v transpose y - v. This is exactly our linear approximation expression.

Remember, that is the gradient of f of x is approximately f of v + grad f of v transpose x - v. Or in other words, f of y is approximately equal to f of v + grad f of v transpose y - v. Y and x are just dummy variables, they do not really matter.

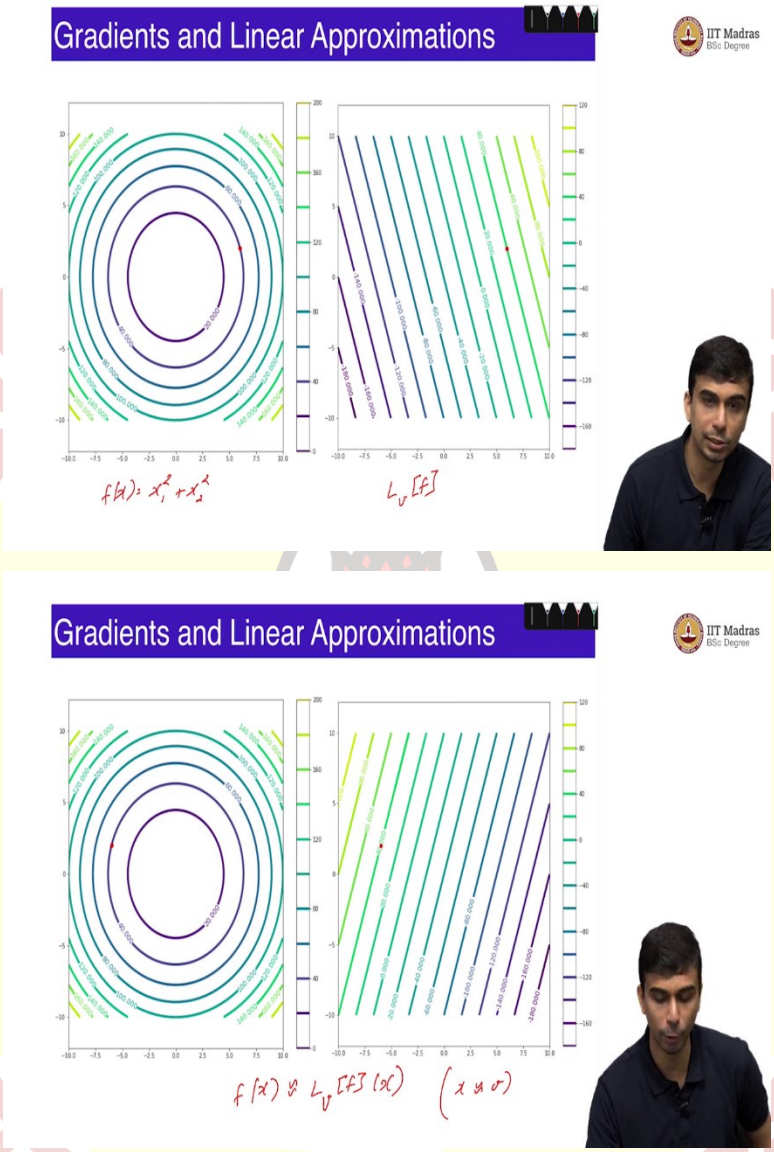
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So, here are a couple of examples to run. So, we will take a simple example first. Let us consider the function f from R2 to R, which is f of x1 , x2 is equal to x1 square + x2 square. Now, here is the question, approximate f around, I want the linear approximation of f around the point let us say 6 , 2. So, v is equal to 6 , 2 in our case. What is f of v first? F of v is equal to 6 square + 2 square which is 40. What is grad f of v? Grad f of v is 2v1, 2v2 which is 12 , 4. This is because grad f of x is 2x1, 2x2, so grad f evaluated at v in this case 2v1, 2v2 which is 12 , 4.

With this we can actually give the answer which is f of x is approximately equal to 40 + grad f of v, which is transposed 12 , 4 × x - v, which is x1 - 6, x2 - 2. We can simplify this further and say 40 + 12 into x1 - 6 + 4 into x2 - 2, which would be 40 + 12x1 + 4x2 - 72 - 8, which would be 12x1 + 4x2 - 40. This is the linear approximation of f, which is x1 square + x2 square around the point 6 , 2. So, this is particularly valid around what point, well, x1 , x2 is approximately 6 , 2. When your vector x is a neighborhood of 6 , 2, this is a valid approximation.

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Let us visualize this further. Here is a contour plot of the function f, which is f of x is equal to x1 square + x2 square. Here is the point 6 , 2. And on the right you have the contour plot of the function, Lv of f, which we just derived here, which is 12x1 + 4x2 - 40. And because this is a linear function, you can see that the contours are all straight lines and they are equally spaced. This is a linear function.

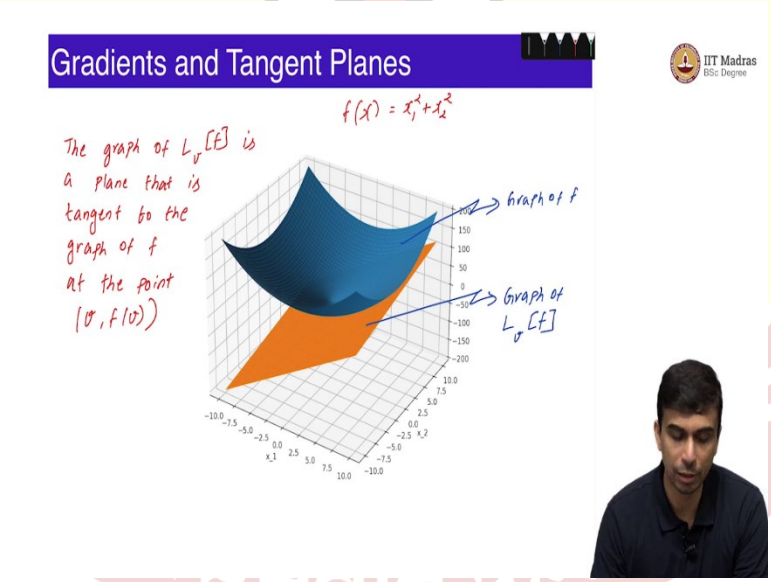
A couple of things to note immediately, which is f of x evaluated at 6 , 2, the value is equal to 40. Consider this circle has value 40. And you can see that the linear approximation of f, are the point 6 , 2 also has the exact value 40. So, the linear approximation is exactly equal to the function value

at the point of approximation. At other points, you do not know, but it is exactly equal at the point of approximation. And there are other properties also. But that is the first property that you should keep in mind.

And the linear approximation does change with the point that you choose. If you want approximate the function f around the point 6 , 2, this is the linear approximation. But if you want to approximate the same function around a point, let us say - 6 , 2, the linear approximation changes. You still have that property that is at the point, you have 40 here and 40 here. The f of v is exactly equal to Lv of f evaluated at v.

At a general point x, this is not true. At the general point x, you only have approximation, f of x, the one to change v to x, this is not exactly equal. This is only approximately equal. And that is also true only when x is approximately equal to v, only when around x equal to v do we have this expression. So, that is the linear approximation viewpoint. And the approximation is completely defined by the gradient.

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And the same way that we have a linear approximation viewpoint for one dimensional functions and tangent line viewpoint, we can have a very similar viewpoint here, which is you can use tangent planes as graphs.

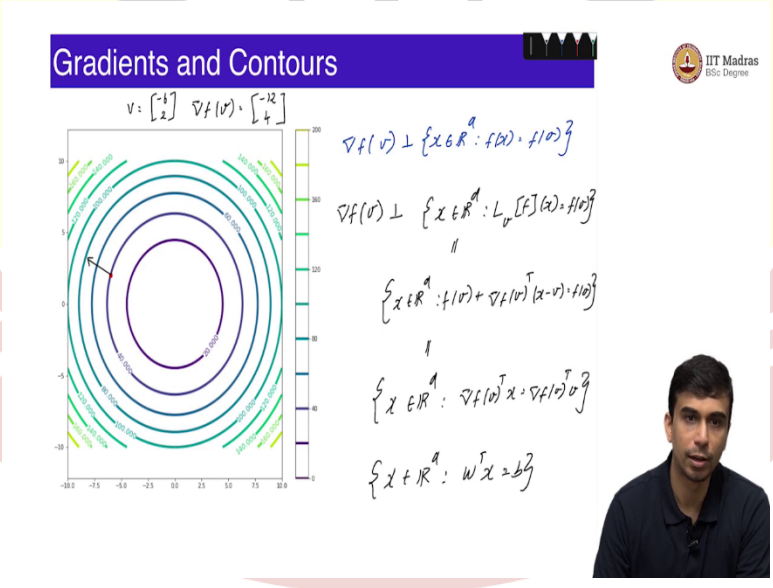
Let us, this is the tangent plane interpretation of the gradient. That is, if you have the graph of the linear approximation of f around the point v, that is the graph of Lv of f, which is going to be a

plane. Recall, graph of a function f from Rd to R is a subset of Rd + 1. Here you have a three dimensional plot of the function f of x is equal to x1 square + x2 square. So, the graph of Lv of f is a plane, because Lv of f is a linear function and linear functions, the graph of linear functions correspond to planes.

A graph of Lv of f is a plane that is tangent to the graph of f at the point v , f of v. So, v is a d dimensional vector and if v of v is a scalar, so this is together is a d + 1 dimensional vectors. So, for example, here you have the blue curve corresponds to the blue surface corresponds to the graph of f. And the orange plane corresponds to the graph of the linear approximation of f around the point v.

You can see that this graph of the linear approximation is a plane which is given and it touches the surface, the blue surface, at a point. And that point is exactly equal to v , f of v. So, this is the other interpretation. That is we have the linear approximation interpretation of gradient, then we have the tangent plane interpretation. The tangent plane is also given by the linear approximation, which in turn is given by the gradient.

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And we have the next interpretation, which uses the contour sets. Here is the statement for the gradient, which as you can see that the gradient of f evaluated at, let us say, at some point v is going to be perpendicular to the contour set given by or the level set given by the set of x in Rd, such that f of x is equal to f of v.

For example, let us say you take the point - 6 , 2 and the gradient of f at that point is simply - 12 , 4. So, what I have done here is I have taken the point - 6 , 2 and I have plotted the vector - 12 , 4 as a vector here. The lengths do not matter, only the direction matters. You can see that the direction, this black arrow is almost perpendicular. It is exactly perpendicular to this level set.

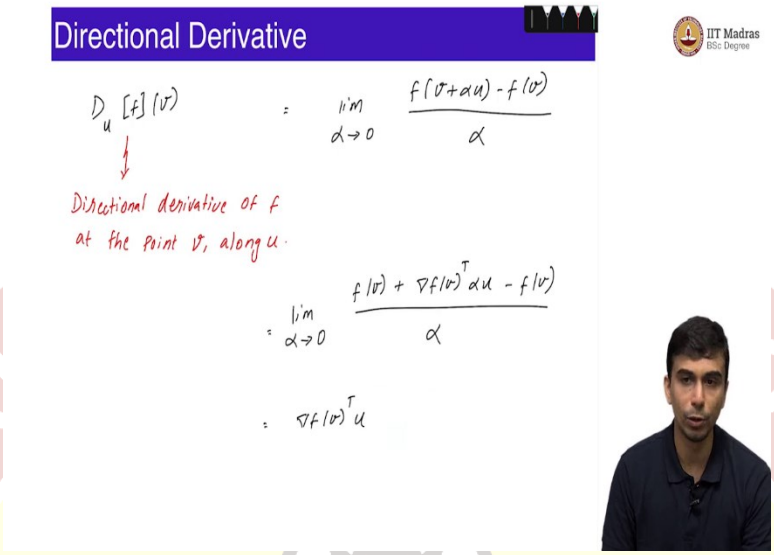
What is this level set? This level set is exactly equal to the set of all x in Rd, such that f of x equal to f of v, which is f of x equal to 40. This is the level set corresponding to f of x equal to f of v, which is f of x equal to 40. And you have the gradient of the function f at v is - 12 , 4, which is perpendicular at that point. And I am not going to prove this completely. But what I am actually going to prove is this statement, which is quite obvious, which is that the gradient of f at v is actually perpendicular to the set x in Rd such that the linear approximation of f evaluated at x is equal to f of v.

So, these two statements are not the same, but, not same as of now, but we will, they are actually the same under certain extra regularity assumptions, but this statement over is very easy to prove. Why is that, because this on the set on the right side is actually a plane. Why is that you have the approximation on f at x is a linear function of x. And this is exactly, planes have a perpendicular vector inbuilt to them. What is that? Well, we will evaluate this set. This set is simply the set of x in Rd such that f of v + grad f of v transpose x - v is equal to f of v.

We will do some moving, term moving around, we get this is the set of x in Rd such that grad f of v transpose x is equal to grad f of v transpose v. And this for the equation of a plane is set of x in Rd such that w transpose x is equal to b. If you have such a plane, you will already know that the vector w is perpendicular to that plane. And you can see that this is exactly of that form grad f of v corresponds to w.

So, grad f of v is actually perpendicular to this plane, which we have a proved now. So, that is the other interpretation of the gradient that is the gradient of a function at a point v is going to be perpendicular to the contour passing through that point.

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Next we move to another useful application or interpretation of the gradient which has to do with directional derivative. What is the directional derivative? We have already seen an example of directional derivative which is, well, we will first say an English redefinition of the directional derivative. We will define directional derivative of the function f at the point v along the direction u. So, you have the function f from Rd to R, u and v are vectors in D dimensional space, u and v are elements in Rd.

So, this is the notation for, I am going to read this as the directional derivative of the function f. I will write this, directional derivative of the function f at the point v along the direction u. This is the, this is how you read the statement. What is the definition of this? Well, the definition is going to be, first create some space here. This is limit α approaching 0, f of v + α u - f of v by α. Let us try to interpret this expression.

What this saying is, we want to move from v to along the direction u. You are at the point v and you want to move along the direction u and you are trying to quantify the rate of change of the function as you move along the direction u. That is the direction and derivative of f along u. How will we compute this thing?

Here, you can use our linear approximations, because α is going to approach 0, you can approximate f of v + α u using linear approximations. This is going to be a limit α approaches 0, f

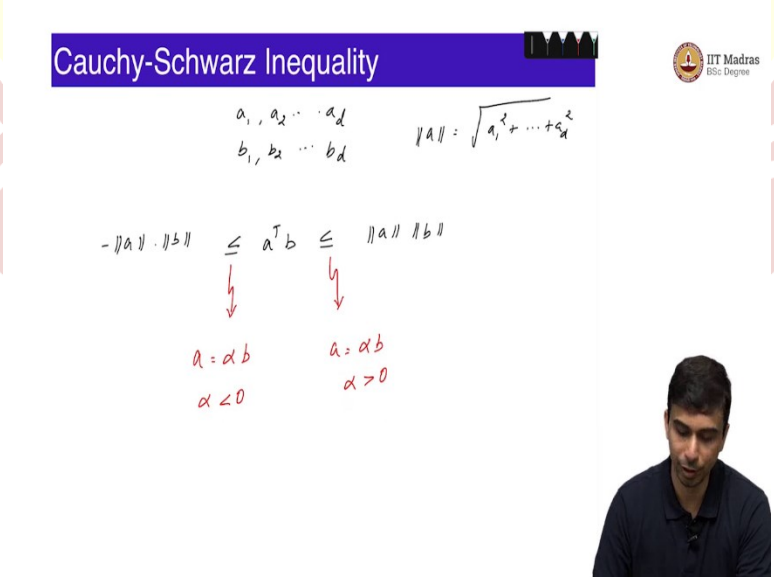
of v + α u is approximately equal to f of v + grad f of v transpose α u - f of v by α. You can see that f of v cancels and α also cancels. So, you are left with grad f of v transpose u.

So, this is nice and particularly interesting because it says that to compute the directional derivative of the function along any direction, all that you need to do is to take the . product of the direction with the gradient. And this is a, in particular, this is nice, because you want to, you are at, let us say, in machine learning, in practice, you often want to minimize a function, which means that you are most approaches take an iterative thing where you are at the current iterate, let us say v, and you want to decide which direction to move.

And you want, you are going to move to new it new iterate from v to let us say v dash, but you want to decide which direction to go. A very natural notion is you are at, you can choose to move at the direction, which decreases the function the most or increases the function the most if you want to maximize it. So, the notion that you are looking for is to what direction to move so that you can increase or decrease the function the most. And the directional derivative is a very important tool for doing that.

We will actually attack that next. And before we do that, however, we will take a slight detour to attack to, actually give a very basic inequality known as the Cauchy-Schwarz inequality, which is useful for controlling the inner product of two vectors grad f of v transpose u.

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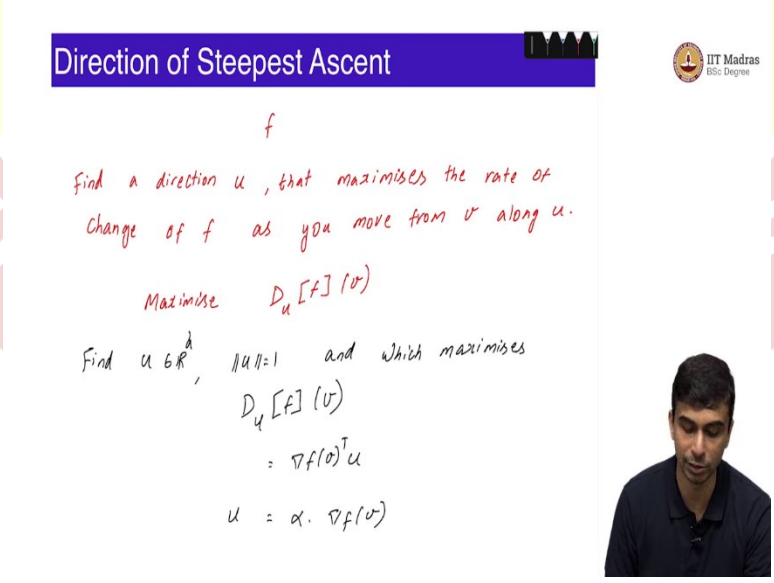


What is that? Well, you have, let us say you have two vectors a and b, whose, they are d dimensional vectors, their components are a1, a2, ., ., . ad, and similarly, b1, b2, ., ., ., bd. You have two d dimensional vectors a and b, and you want to be able to control their inner product. You want to be able to control a transpose b. What we can show is a transpose b is upper bounded and lower bounded by these two terms, which is upper bounded by norm a × norm b and lower bounded by - norm a into norm b.

This norm is the standard Euclidean norm that we have already defined, which is norm a is simply square root of a1 square + a2 square + ., ., . ad square. So, a transpose b is upper bounded and lower bounded between these two terms. And you also happen to know that when does this equality happen. You want to know when this equality happens. And there is a very clear characterization for that.

This equality happens when a is equal to α × b, where α is a scalar, which is negative. If you have a negative scalar α and a is equal to α × b, then you have a transpose b is exactly equal to - norm a, norm b. And when does this equality happen? Well, this equality also happens when a is equal to α b, but α is a positive scalar. So, in other words, if you want to maximize the term a transpose b, you will have, you will choose b is equal to, a is equal to α b and α to be greater than 0.

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And this will all lead us to the next question here, which is, let me clean this. Now you want to find the direction of steepest ascent. Let us say now you are trying to maximize a function f, let us

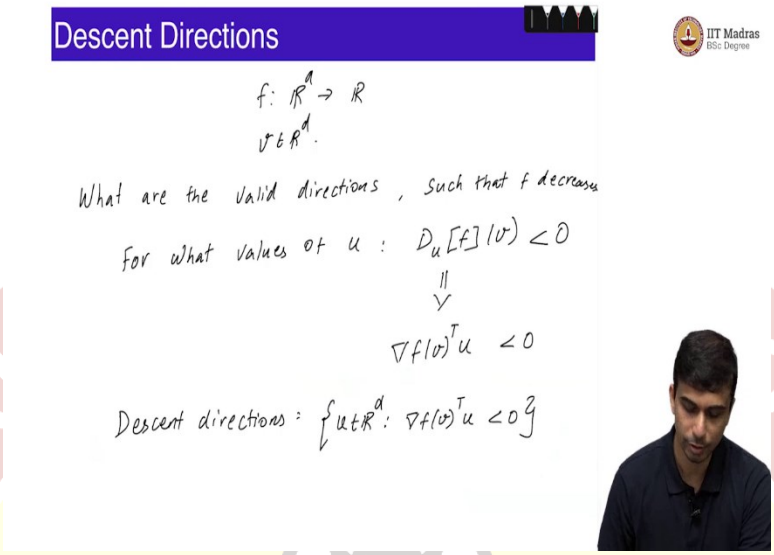
say you have a function f, and you want to maximize it. You are at a point current detractor is v. And you want to choose a direction, find a direction u that maximizes the directional derivative. You want to maximize the rate of change of f as you move from v along u.

In other words, you want to maximize the directional Du of f evaluated at v. Find u which maximizes this term. That is your goal. And we will do a little bit more formally now. What is the direction? A direction cannot have arbitrary length, so we will fix the direction to be a unit norm.

So, find u in Rd such that norm u is equal to 1 and which maximizes the directional derivative of f evaluated at v. But we already know this is exactly equal to grad f of v transpose u. So, v is fixed. The current point is fixed. The only thing that is not fixed is the direction along with u move. So, you want to find a u which maximizes grad f of v transpose u. And that is already given by the Cauchy-Schwarz inequality, which means that you should pick u to be a scalar multiple, a positive scalar multiple of grad f of v.

And you already know that u is a unit norm vector, so α is simply going to be 1 by norm of grad f of v. So, this gives the other interpretation of the gradient which is the gradient is simply the direction of the steepest ascent. If you are at a point v and you want to move along the direction which increases the function the fastest, you would choose the gradient. So, that is the other interpretation, which is the direction of steepest ascent.

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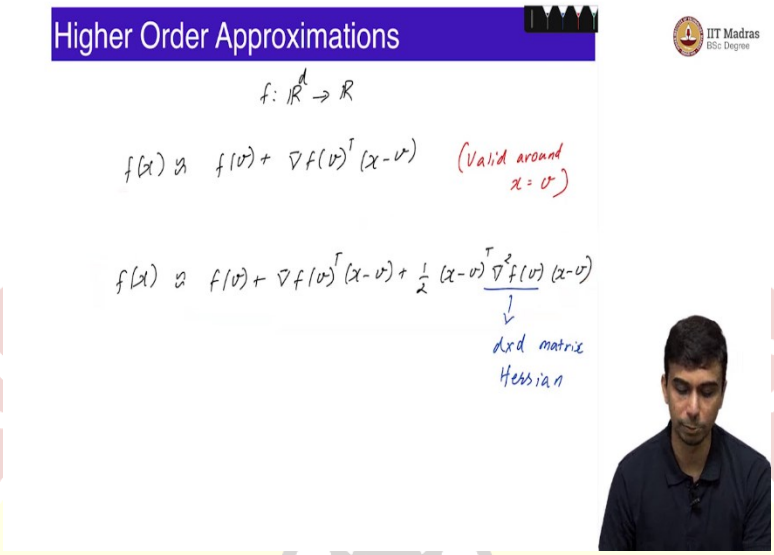


We can finally wrap this up by saying you are at a point let us say v. You have a function f from Rd to R. You are at a point v in Rd. Now, let us say you want to minimize this function. What all directions can you go, what are the valid directions, such that f decreases, or in other words, for what values of u then you say the directional derivative of f along u evaluated at v is less than 0 or in other words, this is grad f of v transpose u less than 0.

So, the set of descent directions is simply the set of u in Rd such that grad f of v transpose u is less than 0. So, this is another tool that you will need when you do optimization that is the set of directions along which you can move so that the function value decreases from the current value. So, that is your descent direction. These are all descent directions.

With that, we will wrap up the alternate grid interpretations. And there are some simple, I mean, there are some slightly advanced rules which I will briefly skim over. We will not go into great detail on these, because we will not be using most of these, but we will do that anyway.

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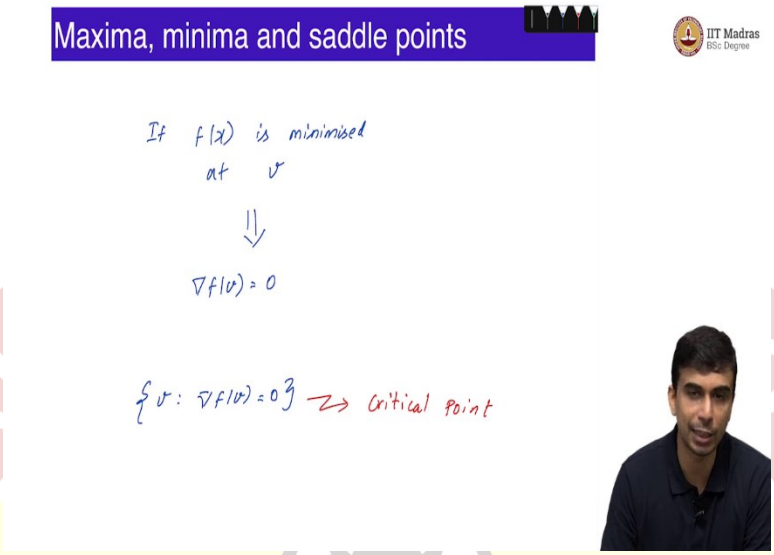
The same way that you did for functions f from R to R, you had linear approximations and you also had higher order quadratic approximations. You can do a very similar thing for functions f from Rd to R also as follows. You have a function f from Rd to R, the linear approximation f of, is simply given by f of x is approximately f of v + grad f of v transpose x - v. This is valid around x is equal to v. This is the linear approximation.

You can do a higher order approximation the same way that you did for one dimensional functions or this is little bit more complicated, which is you can also say f of x is approximately equal to f of v +, the first term looks the same, grad f of v transpose x - V. The second term on the other hand is half x - v transpose, well, the hessian of f evaluated at v into x - v. I will not go into great detail into expanding the second term, but it is analogous to the quadratic term that we had for one dimensional functions.

That is, you have f of x, this is exactly equal to the linear approximation and the extra term, you have this grad square f of v which is called the hessian. This term is a d cross d matrix called the hessian and x - v is a one cross v vector, x - v transpose is one cross v vector and x - v is a d cross one vector together this is a scalar.

I will not go into great detail on this, but you might find more advanced optimization algorithms use a quadratic approximation than the linear approximation, but most of machine learning you will be sticking to linear approximations which is this.

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And the same way that you had maxima, minima and saddle points for one dimensional functions based on the derivative, you have a very similar situation here with gradients and maxima, minima and saddle points that is, if f of x is minimized at some point at, let us say, at some point v, then you have that grad f of v is the zero vector.

But it is not true the other way around, grad f of v equal to 0 does not mean that f of x is minimized at v. But however, if your function is minimized at v then grad f of v is equal to 0, but it is also true for maxima. So, any point where grad f of v equal to 0, the set of v such that grad f of v equal to 0 is called critical points. These points are called critical points. Note however that gradient is a vector. So, this 0 here corresponds to the vector zero not a scalar 0.

For special classes of functions, critical points can be shown to be exactly equal to minima, but in general critical points are not minima, I mean, not guaranteed to be minima or maxima. But in either case critical points are interesting points to analysis. If you want to find the maxima or minima of a function, you can just restrict your attention to points which satisfy this condition. This is called the first order necessary condition for optimality.

And you only focus your attention on critical points and check whether they can be candidate minima or maxima. This is another application of the gradient where you set the gradient to 0 and find interesting points which are critical points and check whether they satisfy your condition. With this, we will wrap up.